

Rohit Agrawal

Data Science & Machine Learning Engineer

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Professional Summary

Aspiring Data Scientist and Machine Learning Engineer with hands-on experience in NLP, predictive modeling, FastAPI backend development, and scalable AI applications. Skilled in building end-to-end ML pipelines, deploying production-ready APIs, and developing explainable AI systems using SHAP and LIME.

Education

- Currently **Bachelor of Technology (Computer Science Engineering)** at Invertis University, Bareilly | 2023 – 2027 | CGPA: 9.0
- Class XII (UP Board)** | 2023 — 92%

Skills

- Languages & Databases:** Python, SQL, PostgreSQL, MySQL, MongoDB, Java
- Data Science & Analytics:** Data Cleaning, Data Preprocessing, EDA, Statistical Analysis, Hypothesis Testing, Feature Engineering, Data Visualization, Predictive Analytics
- Machine Learning & AI:** Scikit-learn, TensorFlow, XGBoost, NLP, Classification, Regression, Clustering, Model Evaluation, Hyperparameter Tuning, SHAP, LIME
- Python Libraries & Visualization:** Pandas, NumPy, PySpark, Matplotlib, Seaborn, Plotly
- Backend & Deployment:** FastAPI, REST APIs, JWT Authentication, Model Deployment, Docker, CI/CD Basics, API Testing (Postman, Swagger), Cloud Basics (AWS/GCP)
- Tools & Platforms:** Git, GitHub, Power BI, Jupyter Notebook, VS Code, Google Colab

Experience / Internship

Python Backend Developer Intern – MakaanMitra

Jan 2026 – Present

- Designed and deployed **25+ scalable REST APIs** supporting tenant & room management system.
- Implemented JWT-based authentication and role-based access control for secure access .
- Integrated PostgreSQL database and optimized queries for efficient data retrieval.
- Reduced API response time by **30%** through query optimization and efficient backend design.
- Collaborated using Git and followed structured development workflows .

Machine Learning & Data Science Intern – AICTE EduSkills (Ministry of Education)

Jan 2026 – Mar 2026

- Completed a 10-week AICTE EduSkills (Ministry of Education) internship focused on ML and data science using real-world datasets.
- Performed EDA, built and optimized ML models, and implemented end-to-end pipelines from preprocessing to evaluation.

Data Science Virtual Intern – EduSkills (Supported by Altair & AICTE)

Oct 2025–Dec 2025

- Conducted EDA on 10k+ records to uncover patterns and support data-driven insights.
- Built and evaluated 3+ ML models (up to 85% accuracy) and created visualizations using Matplotlib and Seaborn.

AI/ML Virtual Intern – EduSkills (Supported by Google)

Apr 2025- Jun 2025

- Implemented end-to-end ML workflows including preprocessing, model training, and evaluation.
- Gained hands-on experience solving real-world AI problems using structured datasets.

ITI – ICTSM (Information & Communication Technology System Maintenance)

Aug 2023–Jul 2025

- Performed system maintenance, OS installation, and basic networking troubleshooting.
- Supported hardware diagnostics and system optimization tasks.

Projects

CareerPilotAI — AI Resume Screening & Career Intelligence Platform

- Developed an AI-powered recruitment platform for resume screening, ATS scoring, candidate ranking, and recruiter workflow automation using NLP, ML, and LLMs.
- Built a hybrid ATS engine combining Sentence-BERT semantic similarity and TF-IDF keyword matching, achieving ~92% candidate-job matching accuracy.
- Implemented intelligent resume parsing, skill-gap analysis, and AI-driven recommendations for automated candidate evaluation.
- Designed recruiter dashboard with candidate shortlisting, GitHub profile insights, resume ranking, analytics, and ATS report generation.
- Developed scalable FastAPI backend with JWT authentication, REST APIs, MongoDB Atlas integration, Docker deployment, and FTP-based resumes on Hostinger server.
- Integrated AI mock interview system with real-time integrity detection using MediaPipe FaceMesh, TensorFlow.js, and face-api.js.

Tech: Python, FastAPI, NLP, Sentence-BERT, TF-IDF, scikit-learn, React, MongoDB Atlas, JWT, Docker, TensorFlow.js, MediaPipe, FTP, Hostinger, LLMs

GitHub: <https://github.com/agrawalrohit937/Resume-Screening-System>

Live Demo: <https://resume-screening-system-lyart.vercel.app/>

Heart Attack Risk Prediction (ML | Deep Learning | Federated Learning)

- Developed an end-to-end machine learning pipeline achieving ~90% accuracy on UCI heart disease dataset.
- Compared 6+ models including **Logistic Regression, SVM, KNN, Random Forest, XGBoost, and DNN**.
- Applied advanced hyperparameter tuning using **GridSearchCV, RandomizedSearchCV, and Optuna**.
- Implemented **Explainable AI (SHAP, LIME)** to interpret predictions and improve transparency.
- Deployed the model using FastAPI and integrated with **Power BI dashboard** for visualization.

Tech: Python, scikit-learn, XGBoost, TensorFlow, SHAP, LIME, Optuna, Power BI.

GitHub: <https://github.com/agrawalrohit937/OriginalNotebook>

Live Demo: <https://heart-attack-prediction-dgzu.onrender.com/>

Participation & Achievements

- IEEE Student Member
- Built and deployed 10+ end-to-end ML applications with live demos
- Achieved ~90% accuracy in predictive modeling project
- Improved backend performance by reducing API response time by 30%
- Selected for Smart India Hackathon Internal Round

Certifications

- Google AI-ML Virtual Internship** — EduSkills
- Altair Data Science Master Internship** — EduSkills
- Python for Data Science** — IBM (Coursera)
- Human Research – Data or Specimens Only Research (CITI Program), MIT Affiliates** | Jan 2026